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Evgeny Sobolev 09.02.1984 y.b.

$y(x) := \sqrt{x}$ $\quad Max32 := \left(2^{32} - 1\right)$ $\quad clz(x) := \text{if } x = 0\ 32$ else $32 - \text{trunc}\left(\log_2\left(\text{trunc}(x) + \frac{Max32}{2^{32}}\right) + \frac{Max32}{2^{32}}\right)$

$msb(x) := 32 - clz(x)$

$dY_32 := \text{trunc}\left(\left(\left(\sqrt{2}\right)^{32} - \left(\sqrt{2}\right)^{31}\right) \cdot 2^{16}\right)$ $\quad Y0_32 := \text{trunc}\left(\sqrt{2} \cdot 2^{15} \cdot 2^{16}\right)$

$dY_31 := \text{trunc}\left(\left(\left(\sqrt{2}\right)^{31} - \left(\sqrt{2}\right)^{30}\right) \cdot 2^{16}\right)$ $\quad Y0_31 := \text{trunc}\left(2^{15} \cdot 2^{16}\right)$

$dY_32 = 1257966796$ $\quad dY_31 = 889516851$

$Y0_32 = 3037000499$ $\quad Y0_31 = 2147483648$

$dY(x) := \text{if } \text{trunc}\left(\frac{clz(x)}{2}\right) \cdot 2 = clz(x)\ dY_32$ else dY_31

$Y0(x) := \text{if } \text{trunc}\left(\frac{clz(x)}{2}\right) \cdot 2 = clz(x)\ Y0_32$ else $Y0_31$

$xScaled(x) := \text{trunc}\left(\text{trunc}(x) \cdot 2^{clz(x)+1} - 2^{32}\right)$

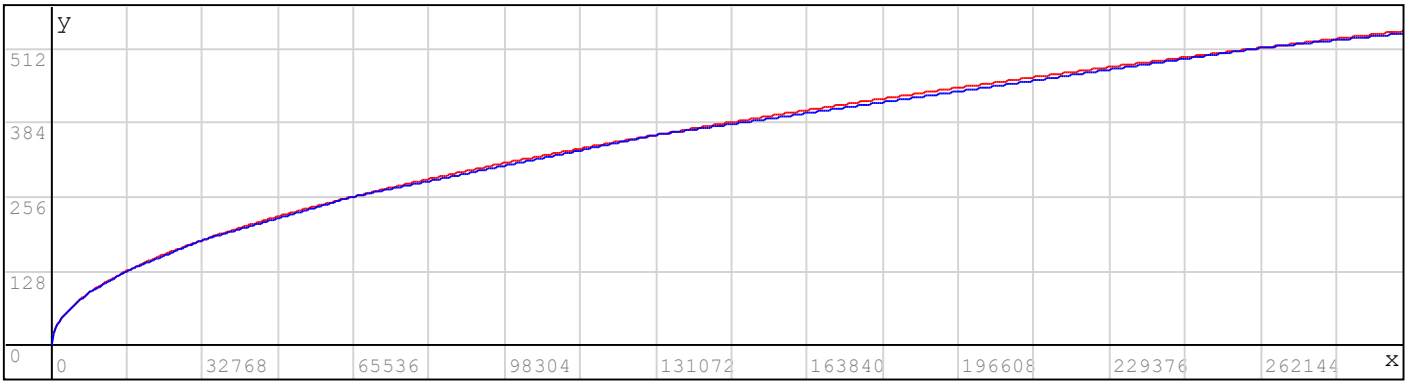
$sqrtLinearScaled(x) := \text{trunc}\left(\frac{\text{trunc}(xScaled(x) \cdot dY(x))}{2^{32}}\right) + Y0(x)$

$sqrtLinear(x) := \frac{sqrtLinearScaled(x)}{2^{\text{trunc}\left(\frac{clz(x)}{2}\right)}}$

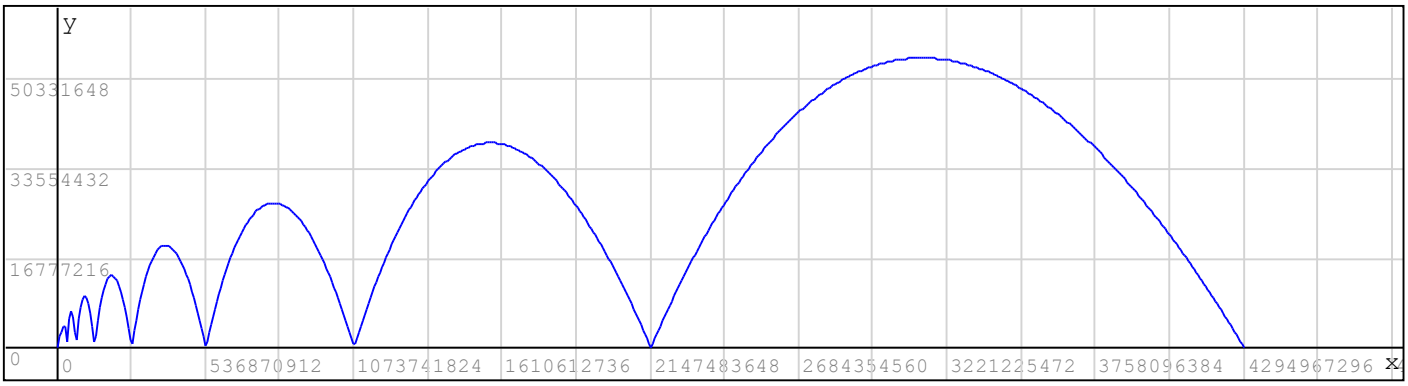
$sqrtLinearErr(x) := y(x) \cdot 2^{16} - sqrtLinear(x)$

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$$\begin{cases} \dfrac{sqrtLinear(x)}{2^{16}} \\ \sqrt{x} \end{cases}$$



$$\left\{ \sqrt{x} \cdot 2^{16} - sqrtLinear(x) \right.$$

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$$xMaxErr_32 := \operatorname{trunc}\left(\frac{2^{32}}{\left(4-2\cdot\sqrt{2}\right)^{2}}\right) \qquad xMaxErrScaled_32 := \operatorname{trunc}\left(\frac{2\cdot\sqrt{2}-1}{4}\cdot 2^{32}\right)$$

$$yMaxErr_31 := \operatorname{trunc}\left(\sqrt{\frac{xMaxErr_32}{2}}\cdot 2^{16} - sqrtLinear\left(\frac{xMaxErr_32}{2}\right)\right)$$

$$yMaxErr_32 := \operatorname{trunc}\left(\sqrt{xMaxErr_32}\cdot 2^{16} - sqrtLinear\left(xMaxErr_32\right)\right)$$

$$xMaxErr_32 = 3129112985 \qquad yMaxErr_31 = 38154242 \qquad yMaxErr_32 = 53958246$$

$$xMaxErrScaled_32 = 1963258675$$

$$yMaxErrScaled(x) := \text{if } \operatorname{trunc}\left(\frac{clz(x)}{2}\right)\cdot 2 = clz(x) \quad yMaxErr_32 \text{ else } yMaxErr_31 \qquad \frac{yMaxErr_32}{65536} = 823.33749$$

$$xMaxErr(x) := \frac{xMaxErr_32}{2^{\operatorname{trunc}(clz(x))}} \qquad yMaxErr(x) := \frac{yMaxErrScaled(x)}{2^{\operatorname{trunc}\left(\frac{clz(x)}{2}\right)}}$$

$$dXMaxErr(x) := \left|\left(xScaled(x) - xMaxErrScaled_32\right)\cdot 2\right|$$

$$dXMaxErr(x) := \text{if } xScaled(x) < xMaxErrScaled_32 \left|\left(xMaxErrScaled_32 - xScaled(x)\right)\right|\cdot\frac{Max32}{xMaxErrScaled_32} \text{ else } \left|\left(xScaled(x) - xMaxErrScaled_32\right)\right|\cdot\frac{Max32}{Max32 - xMaxErrScaled_32}$$

$$dQMaxErr(x) := \operatorname{trunc}\left(\frac{\left(dXMaxErr(x)\right)^{2}}{2^{32}}\right) \qquad dXMidErr(x) := \left|2^{31} - xScaled(x)\right|\cdot 2$$

$$dQMaxErrInv(x) := 2^{32} - dQMaxErr(x) \qquad dQMidErr(x) := \operatorname{trunc}\left(\frac{\left(dXMidErr(x)\right)^{2}}{2^{32}}\right)$$

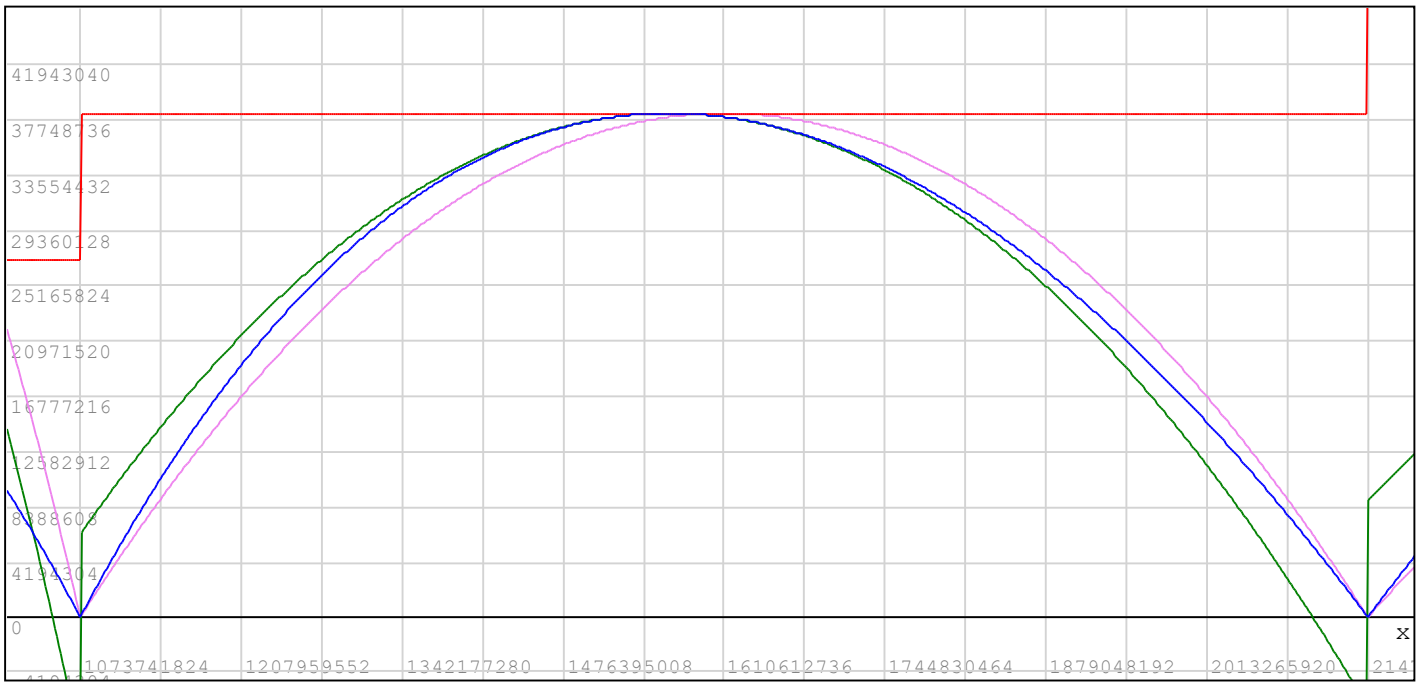
$$parMaxErr(x) := \operatorname{trunc}\left(\frac{dQMaxErrInv(x)\cdot yMaxErrScaled(x)}{2^{32}}\right) \qquad dQMidErrInv(x) := 2^{32} - dQMidErr(x)$$

$$parMidErr(x) := \operatorname{trunc}\left(\frac{dQMidErrInv(x)\cdot yMaxErrScaled(x)}{2^{32}}\right)$$

$$par(x) := \left(\operatorname{trunc}\left(\frac{parMaxErr(x)\cdot dQMidErrInv(x)}{2^{32}}\cdot(1+0)\right) + \operatorname{trunc}\left(\frac{parMidErr(x)\cdot dQMidErr(x)}{2^{32}}\cdot(1.036486)\right)\right)$$

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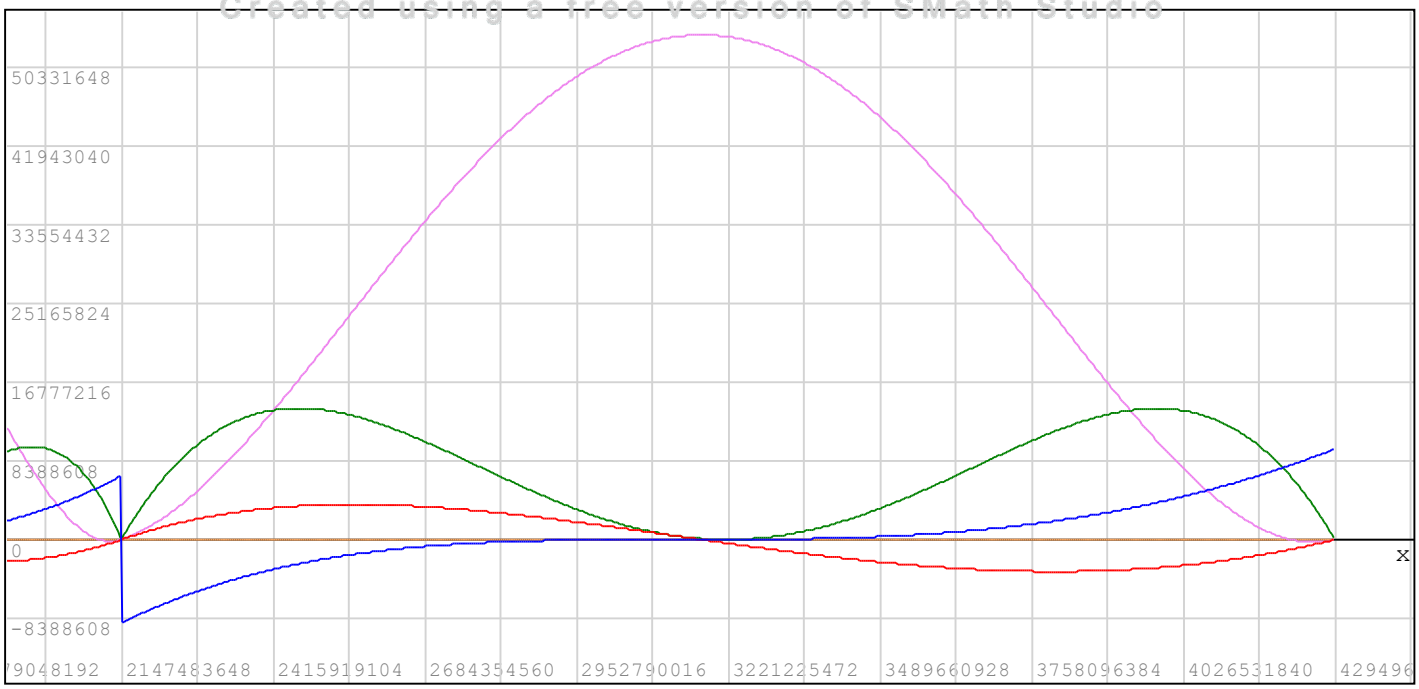
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$$\begin{cases} sqrtLinearErr(x) \\ yMaxErr(x) \\ parMaxErr(x) \\ parMidErr(x) \end{cases}$$

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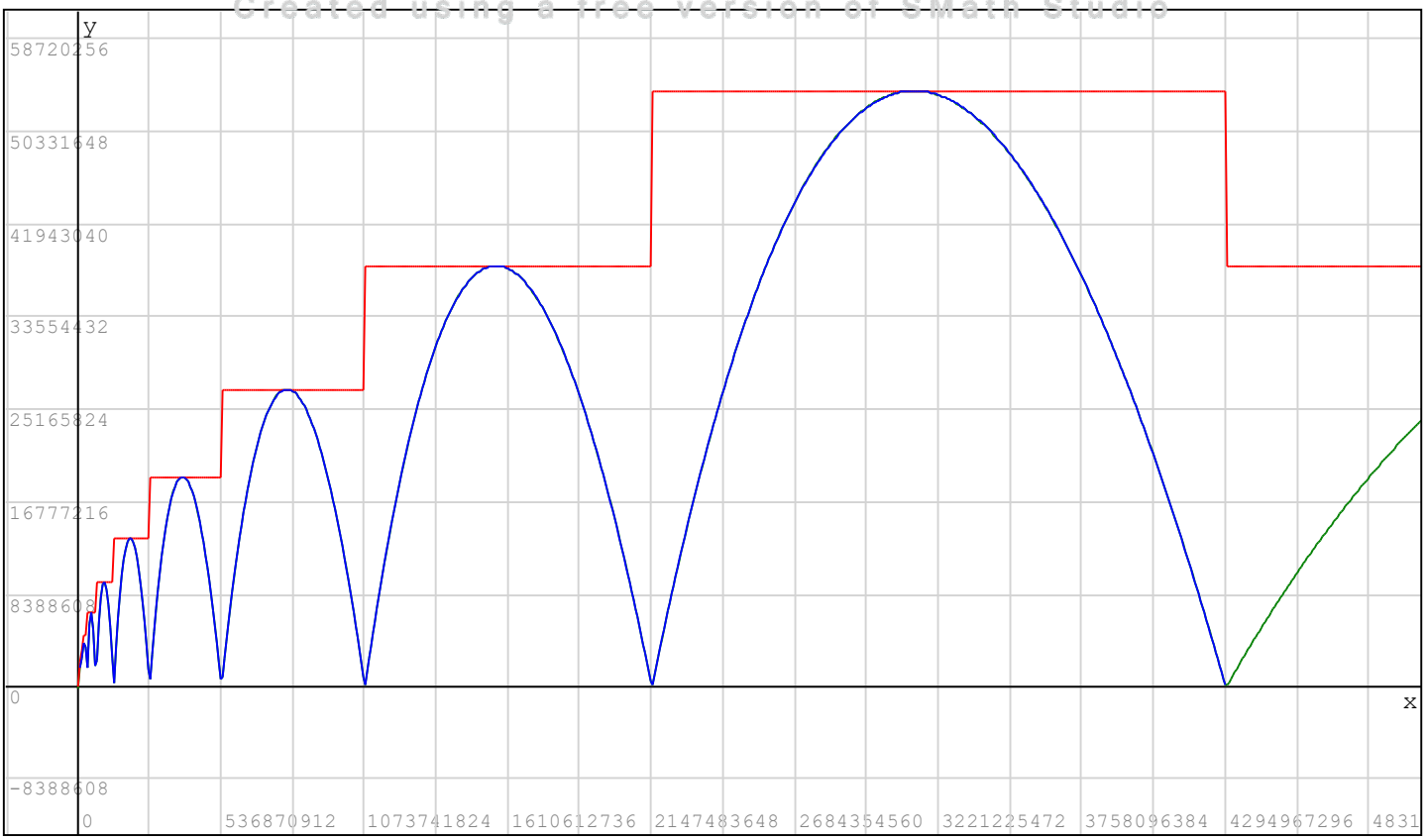
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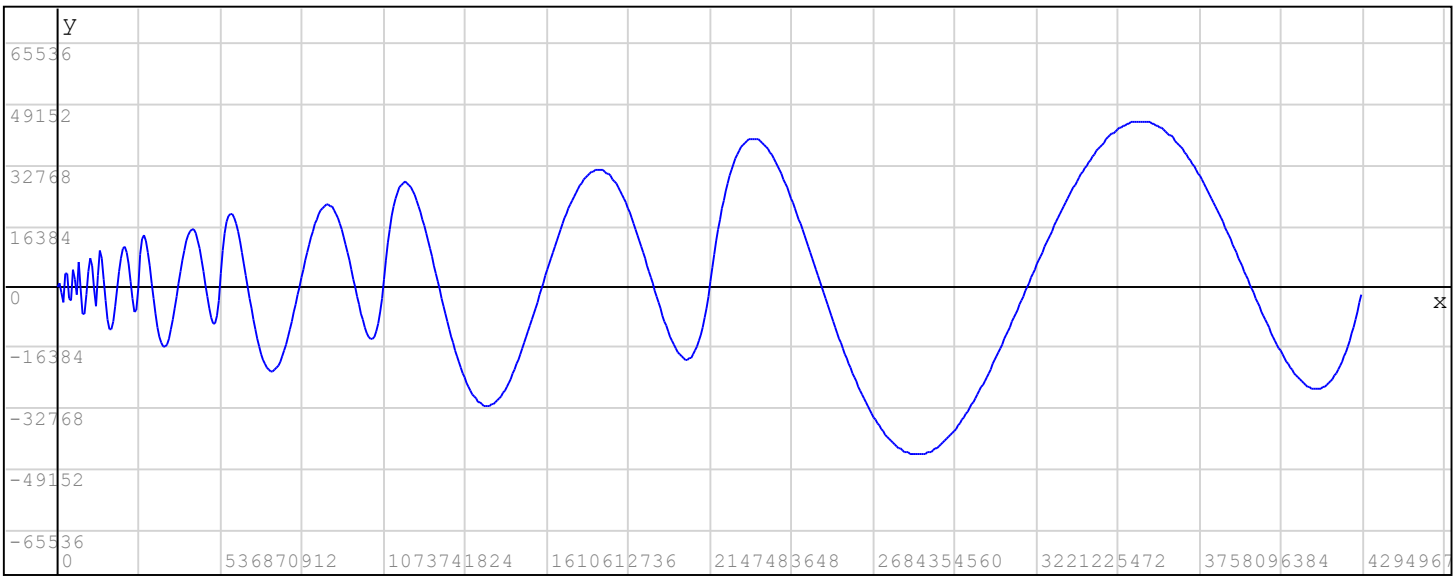
$$\begin{cases} sqrtLinearErr(x) - parMaxErr(x) \\ sqrtLinearErr(x) - parMidErr(x) \\ sqrtLinearErr(x) - parMaxErr(x) \cdot \dfrac{dQMidErrInv(x)}{2^{32}} \\ sqrtLinearErr(x) - parMidErr(x) \cdot \dfrac{dQMidErr(x)}{2^{32}} \\ sqrtLinearErr(x) - par(x) \end{cases}$$

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$$\begin{cases} sqrtLinearErr(x) \\ yMaxErr(x) \\ \dfrac{par(x)}{2^{\operatorname{trunc}\left(\frac{clz(x)}{2}\right)}} \end{cases}$$



$$\begin{cases} y(x)\cdot 2^{16-\frac{sqrtLinearScaled(x)+par(x)}{2^{\operatorname{trunc}\left(\frac{clz(x)}{2}\right)}}} \end{cases}$$

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